A Compact 2–19.5-GHz 150-ps True Time Delay Circuit With High Area Efficiency of 882.4-ps/mm^2

Yanting Chen, Xiangyu Meng, *Member, IEEE*, Weijia Qiu, Chaojie Zeng, Junbin Ouyang, Yuwen Tao, *Student Member, IEEE*, and Baoyong Chi, *Senior Member, IEEE*

***Abstract*—In this brief, a novel switched true-time-delay (TTD) circuit is presented using a 65-nm CMOS process. It employs cascaded series-capacitor all-pass networks (SCAPNs) for delay compensation, along with a novel layout based on an 8-shaped transformer, to construct a compact and broadband delay unit. The design leverages the self-canceling coupling property of an 8-shaped transformer to integrate the inductors required for both APN stages into a single component, achieving an ultra-compact footprint. To verify the proposed structure, a 5-bit switched-line TTD is constructed using two single-pole double-throw (SPDT) and four double-pole double-throw (DPDT) switches. Experimental results show that, with a core size of only 0.209 × 0.816 mm^2, the proposed TTD circuit achieves a maximum delay tuning range of 150 ps and a minimum delay step of 4.838 ps from 2 to 19.5 GHz, while maintaining RMS delay error below 3.7 ps. The corresponding area efficiency of 882.4 ps/mm^2 represents a record high for switched-line TTDs.**

***Index Terms*—All pass network (APN), 8-shaped transformer, CMOS, group delay, true time delay (TTD).**

I. Introduction

Ultra-wideband phased arrays require true-time-delay (TTD) circuits to overcome the fundamental beam-squinting limitation inherent in conventional phase shifters [1], [2]. Consequently, TTD technology has become pivotal for advanced beamforming. However, its implementation remains challenging, as it must concurrently satisfy stringent demands for wide bandwidth, high precision, compact size, and low loss.

Active delay lines, such as Gm-C circuits [3], [4], [5], offer compact size and large delay but suffer from inherently narrow bandwidth, poor linearity, and high power consumption. N-path circuits [6], [7] provide an alternative approach, but their maximum operating frequency is strictly climited by the clock sampling rate. Switched-line TTDs [8], [9], [10], [11], [12], [13] have emerged as a promising path to achieve ultra-wideband operation with zero DC power and excellent linearity. However, conventional switched-line TTDs suffer from high insertion loss and occupy a large area, which results in poor area efficiency.

This brief presents a compact 5-bit switched-line TTD circuit that overcomes the size limitations. The design employs a delay compensation scheme based on cascaded coupled series-capacitor all-pass networks (SCAPNs). An area-minimizing 8-shaped transformer layout is utilized to construct the broadband delay cell. This implementation achieves a 150 ps delay tuning range from 2 to 19.5 GHz, a high area efficiency of 882.4 ps/mm^2, and low RMS delay error.

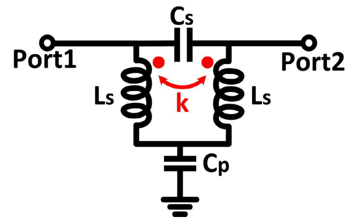

Fig. 1. Schematic of the SCAPN TTD unit.

II. Design of the Proposed Compact TTD Unit

A. Theory of the Proposed Delay Unit

Fig. 1 depicts the the schematic of series-capacitor all-pass network (SCAPN) TTD unit. The two inductors in SCAPN are mutually coupled with a coefficient k. This type of APN is capable of generating a transfer function with all-pass characteristics in magnitude and a tunable group delay. The design equations for achieving full-band perfect impedance matching in such an APN can be obtained as [14]

$$k = -\frac{1-Q^2}{1+Q^2} \quad L_s = \frac{(Q^2+1)Z_0}{2Q\omega_r} \tag{1}$$

$$C_s = \frac{1+k}{2Z_0^2}\times \quad L_sC_p = \frac{2(1-k)}{Z_0^2}\times L_s \tag{2}$$

where Q and ω_r are the design parameters that can be flexibly changed to obtain different group delay responses, and Z_0

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Yanting Chen, Xiangyu Meng, Weijia Qiu, Chaojie Zeng, Junbin Ouyang, and Yuwen Tao are with the School of Electronics and Information Technology, Sun Yat-sen University, Guangzhou 510275, China (e-mail: mengxy26@mail.sysu.edu.cn).

Baoyong Chi is with the School of Integrated Circuits, BNRist, Tsinghua University, Beijing 100084, China.

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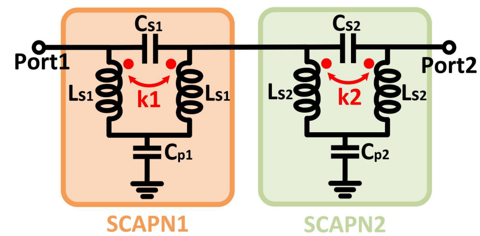
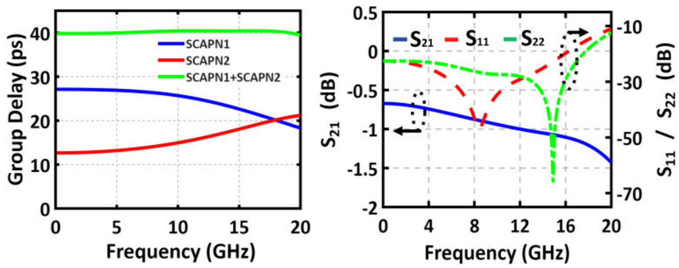


Parameter	Value
Ls1(pH)	430
k1	-0.55
Cp1(fF)	550
Cs1(fF)	50
Ls2(pH)	210
k2	-0.4
Cp2(fF)	268
Cs2(fF)	65

Fig. 2. Schematic of the proposed delay unit and a numerical design example for achieving a total delay of 40 ps.



(a) (b)

Fig. 3. Simulated results of the proposed delay unit for achieving a total delay of 40 ps.

represents the characteristic impedance. And the group delay of SCAPN can form as [14]

$$T_{gd} = \frac{2Q\omega_r(\omega^2+\omega_r^2)}{Q^2(\omega^2-\omega_r^2)^2+\omega^2\omega_r^2} \quad T_{gd} = \frac{2Q(\omega^2+1)}{Q^2(\omega^2-1)^2+\omega^2} \tag{3}$$

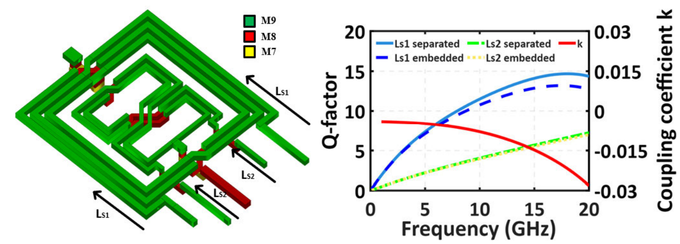
where ω represents the angular frequency. By normalizing the expression such that $\omega_r = 1$, it can be simplified to

$$\omega(T_{gd,max}) = \omega_r\sqrt{\sqrt{4-\frac{1}{\mathrm{Q}^2}}-1} \tag{4}$$

$$T_{gd,max} = \frac{4Q}{\omega_r} \quad T_{DC} = \frac{2}{Q\omega_r} \tag{5}$$

As can be seen from Eq (3), when $Q < 0.577$, the maximum group delay occurs at DC. As Q increases, the frequency corresponding to this maximum delay approaches ω_r, as described by Eq (4) and (5). Therefore, adopting $Q > 0.577$ enables compensation for the delay roll-off of conventional SCAPN structures at higher frequencies. The key design parameters include Q, ω_r, and the target delay time. A higher Q in the second stage provides greater compensation capability, but it must be carefully matched to the delay roll-off slope of the first stage, ensuring that the two stages exhibit opposite yet equal slopes. However, a higher Q also results in lower group delay at low frequencies and imposes bandwidth limitations, necessitating a trade-off between Q and ω_r.

As shown in Fig. 2, schematic of the proposed delay unit and a numerical design example for a total delay of 40 ps within 20 GHz is presented. To minimize the overall area of the two-stage transformer, the delay values for the two stages are designed to be 27.5 ps and 12.5 ps, respectively and pole Q of 0.5 and 0.654, respectively, with all parameters listed in Fig. 2. As shown in Fig. 3(a), the two SCAPN stages exhibit complementary frequency-dependent delay responses. Specifically, the delay of the first stage decreases with increasing frequency, while that of the second stage increases. After cascading the two SCAPNs together, a flat delay response at the level of 40 ps can be finally realized. As presented in Fig. 3(b), the circuit exhibits insertion loss better than 1.4 dB and return losses below -10 dB from DC to 20 GHz.



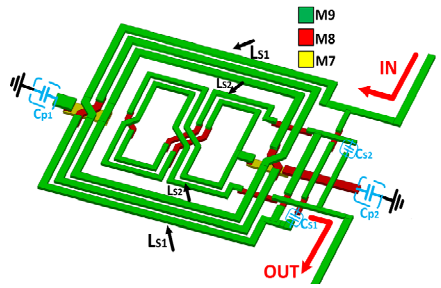
(a) (b)

Fig. 4. (a) 3D-view of the proposed transformers for SCAPNs and (b) simulated coupling coefficients between the coils and Q-factors of inductors L_{s1} and L_{s2} in the cases of seperated and embedded.

B. Compact Layout Design of the Proposed Delay Unit

In the proposed TTD unit, two SCAPNs are cascaded, each requiring a transformer. The corresponding area is largely occupied by these high-inductance transformers. In conventional design methodologies, the transformers or inductors are laid out separately to avoid unwanted mutual coupling, leading to significant circuit size and area overhead. To achieve high area efficiency, this work introduces a novel layout implementation of the delay unit in 65-nm CMOS technology, as illustrated in Fig. 4(a). The proposed layout employs an 8-shaped transformer configuration. As shown, the transformer in the first SCAPN (transformer 1) is constructed from two conventional rectangular inductors in the top metal layer(M9), while the transformer in the second SCAPN (transformer 2) is twisted into an 8-shape. The 8-shaped geometry provides a self-canceling coupling characteristic, enabling transformer 2 to be embedded within the coil of transformer 1 with negligible coupling effect. Fig. 4(b) demonstrates that the coupling coefficient between the two transformers is negligibly small, and that the embedding of inductors L_{s1} and L_{s2} results in only a minor degradation of the Q-factor. Additionally, the Q-factor affects both the insertion loss and the minimum operating frequency.

To achieve a total delay of 40 ps, the two-stage SCAPN is designed with DC delays of 26.86 ps and 12.84 ps for the first and second stages, respectively, and pole Q of 0.49 and 0.67. Fig. 5 illustrates the layout of the proposed area-efficient delay cell along with its design parameters. In the post-layout implementation, parasitic capacitance and reduced inductor Q-factors (relative to pre-layout simulations) lead to increased insertion loss. To preserve the targeted delay flatness and delay magnitude, the values of C_{P1},C_{P2},C_{S1}, C_{S2} were further decreased. Consequently, the S11 peak in the post-layout results shifts to a higher frequency. A layout comparison with the conventional cascaded SCAPN and SIAPN (series-inductor APN) design is shown in Fig. 6, where the



26.86ps				12.84ps			
Ls1(pH)	k1	Cp1(fF)	Cs1(fF)	Ls2(pH)	k2	Cp2(fF)	Cs2(fF)
415.4	-0.61	526.9	26.7	210.2	-0.38	200.5	39.1

Fig. 5. Compact layout of the proposed 40 ps delay unit with key component values.

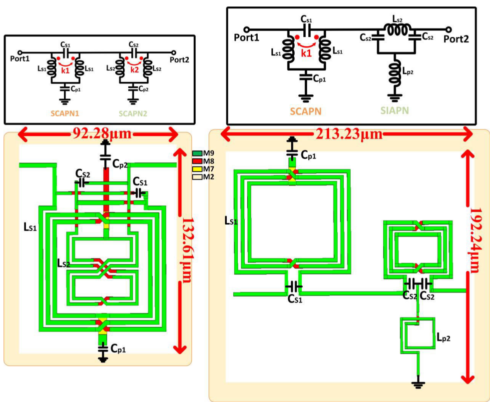


Fig. 6. Layout area comparison: proposed versus conventional cascaded SCAPN and SIAPN.

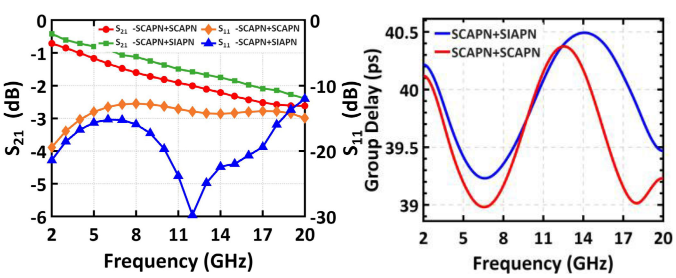
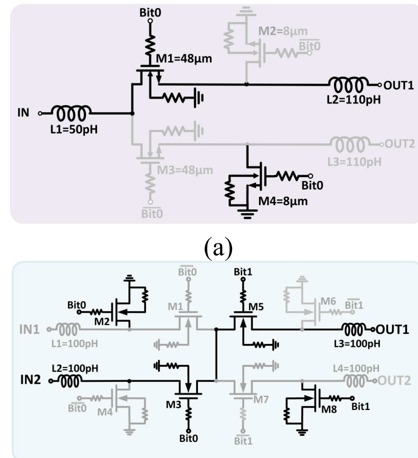


Fig. 7. Simulation results comparison.

SIAPN inductors L_{p2} is embedded inside the coil of inductor L_{s2}. Fig. 7 compares the simulated S-parameters and delay responses between the proposed structure and the conventional cascaded SCAPN and SIAPN design. To facilitate a more comprehensive comparison, Table I summarizes the key performance metrics of this work and the current advanced cascaded SCAPN and SIAPN design [15]. As shown, the proposed circuit achieves a comparable delay of 40 ps within a substantially more compact footprint. Its insertion loss up to 15 GHz is about 0.3 dB higher than that of the advanced cascaded SCAPN and SIAPN design. Additionally, the delay variation is slightly larger by approximately 1%. The area of the advanced cascaded SCAPN and SIAPN design is 0.0276 mm, while the proposed structure occupies 0.0123 mm.



(a)

(b)

Fig. 8. Schematic of (a) SPDT and (b) DPDT switch.

TABLE I

PERFORMANCE COMPARISON OF DIFFERENT TTD UNITS STRUCTURES

Structure	Frequency Range(GHz)	Insertion(dB)	Delay Variation	Circuit Size(mm²)	Area Efficiency(ps/mm²)
Conventional SCAPN+SIAPN	2-20GHz	0.41-2.39	<3.5%	0.0409	878
Advanced SCAPN+SIAPN	2-15GHz	0.31-1.91	<2.5%	0.0276	1449.3
Proposed SCAPN+SCAPN	2-20GHz	0.71-2.62	<3.77%	0.0122	3268.7

The proposed layout achieves a corresponding area efficiency of 3268.7 ps/mm, which is 2.25 times that of its advanced counterpart. Therefore, to obatain a same level of time delay, the proposed layout can achieve a size reduction of up to 55.4%.

C. Design of the Proposed 5-bit Switched-Line TTD

To verify the feasibility of the proposed delay unit, a 5-bit switched TTD with high area efficiency is presented. Fig. 8(a) shows the circuit schematic of the SPDT switch using a series-shunt configuration to improve isolation, where the total width of the series and shunt transistors is chosen to 48 and 8 μm, respectively. Meanwhile, three inductors (L_1 - L_3) are added at the three ports for impedance matching. Furthermore, the DPDT switch is formed by connecting two SPDT switches in back-to-back configuration, as shown in Fig. 8(b). Considering bandwidth, delay time, insertion loss, and delay error, Fig. 9 shows the schematic and layout of the proposed 155 ps switched-line TTD circuit. The circuit has 5-bits controlled by two SPDT switches and four DPDT switches. The least significant bits (LSBs: 5, 10, 20 ps) employ a single SCAPN per delay unit, whereas the most significant bits (MSBs: 40, 80 ps) employ cascaded different numbers of the the proposed 40 ps unit. Meanwhile, an attenuation unit is integrated into the switched delay circuit to balance the insertion loss between the reference and delay states.

III. MEASURED RESULTS

The proposed TTD circuit was fabricated in a standard 65-nm CMOS process. A die micrograph of the fabricated chip

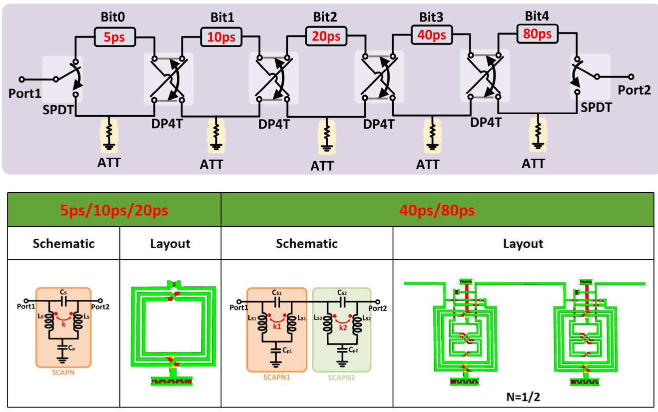


Fig. 9. Block diagram of the designed 155 ps 5-bit CMOS TTD circuit.

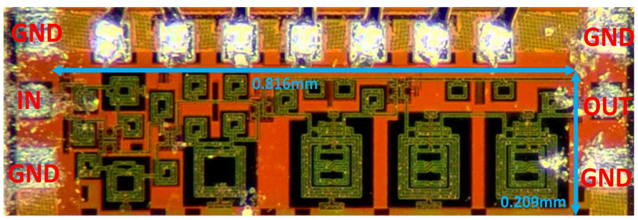


Fig. 10. Die photo of the fabricated TTD chip.

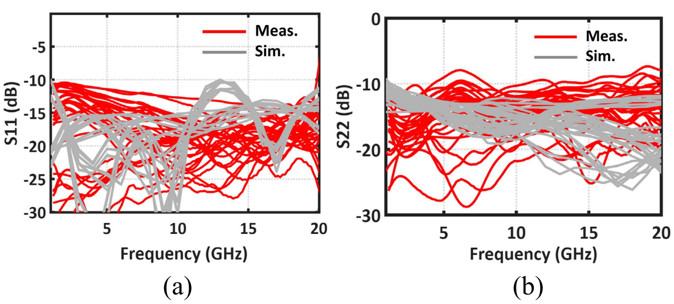


Fig. 11. (a) Simulated and measured S_{11} and (b) S_{22} of the 32 delay states.

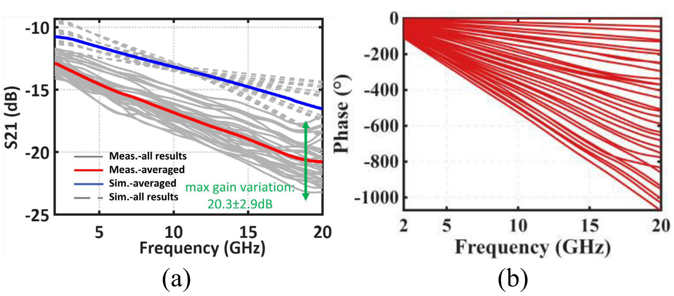


Fig. 12. (a) Simulated and measured S_{21} and (b) measured phase of the 32 delay states.

is shown in Fig. 10, with a core area of 0.209×0.816 mm^2 (pads excluded). Fig. 11 presents the simulated and measured input and output return losses (S11 and S22) across all 32 delay states from 2 to 20 GHz, both of which remain below than −9 dB across the entire band. Fig. 12(a) compares the simulated and measured insertion loss for each state along with the averaged simulated and measured results from 2 to 20 GHz. Over the frequency range of 2-19.5 GHz, the insertion loss variation across all delay states is confined within ± 2.9 dB, with the worst case occurring at 19.5 GHz. The average measured insertion loss ranges from 12.8 dB to 20.3 dB, which is above simulation. This discrepancy results from process-induced increase in switch parasitic resistance and the generally lower Q-factors of the inductors in measurement. Fig. 12(b) shows the measured unwrapped phase of the circuit across all delay states. The phase increases almost linearly with frequency in all states, and its slope grows as the delay setting increases, reaching a relative phase of 1061.7° at 20 GHz. Fig. 13 presents the relative group delay and the averaged relative time delay for all 32 delay states from 2 to 20 GHz. Non-ideal parasitic components inevitably affect both the group delay and the S-parameters in the measurement, which causes larger variation than simulation. The results confirm that the designed TTD circuit successfully provides 5-bit group delay over the 2-20 GHz band. The measured maximum time delay is 150 ps, resulting in an actual delay step of 4.83ps, which is 0.17 ps smaller than in simulation.

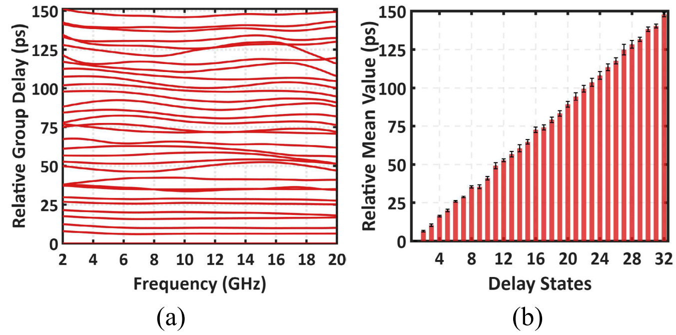


Fig. 13. (a) Measured relative group delay and (b) measured relative mean value of the 32 delay states.

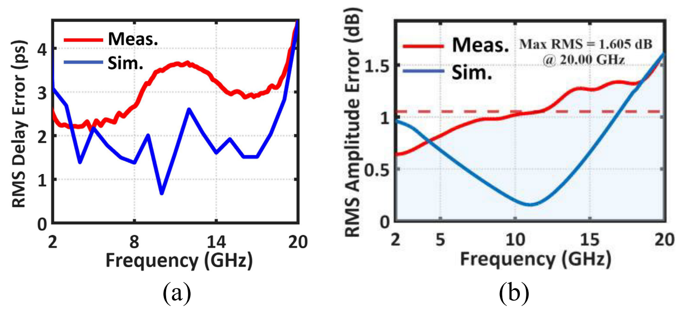


Fig. 14. (a) Simulated and measured RMS delay error and (b) RMS amplitude error.

The simulated and measured RMS delay error and RMS amplitude error are summarized in Fig. 14. As shown, the RMS delay error remains below 3.7 ps from 2 to 19.5 GHz, while the RMS amplitude error is below 1.51 dB, with an average RMS amplitude error of 1.05 dB over 2-20 GHz. The simulated and measured IP_{1dB} is plotted in Fig. 15. The measured $IP1_{dB}$ is better than 5.2 dBm across all states from 2 GHz to 19.5 GHz, a limitation mainly attributed to the junction diodes in the CMOS SPDT and DPDT switches and process. Table II summarizes the performance of the proposed TTD circuit and compares it with other recently reported state-of-the-art designs. As listed in Table II, the proposed TTD achieves an area efficiency of 882.4 ps/mm^2. Over 2-19.5 GHz, the corresponding insertion loss per delay ranges

TABLE II
COMPARISON WITH THE STATE-OF-ART TTD CIRCUIT

Ref.	Tech.	Topology	Max Delay (ps)	Delay Step (ps)	Freq. (GHz)	DBW	IL(dB)	IL/Delay (dB/ps)	Loss-Variation (dB)	Loss-Variation/Delay (dB/ps)	RMS Amplitude Error(dB)	RMS Delay Error(ps)	DC Power(mW)	IP_{1dB} (dBm)	Core Area(mm²)	Area Efficiency (ps/mm²)	FOM
[16]	40 nm CMOS	Trombone	125	17.85	5-23	2.25	4.5	0.036	±1.5	0.012	N/A	N/A	55.8	-2.7	0.85	147.1	588.2
[10]	65 nm CMOS	Switched	400	1.54	5-11	2.4	41	0.1025	±10	±0.025	N/A	<4.5	Passive	11.2	2.46	162.6	23.8
[2]	45 nm SOI CMOS	Switched	21	3	10-24	0.294	8.1	0.39	±4.3	±0.2048	N/A	N/G	Passive	17	0.13	161.5	279.2
[17]	65 nm CMOS	Switched	106.7	4.85	1-11	1.067	2.8-15.5	0.03-0.15	±3.95	±0.037	N/A	<4.6	Passive	11	0.35	304.9	333.2
[18]	65 nm CMOS	Switched	106	2.35	3-20	1.802	5.7-17	0.05-0.16	±2	±0.0189	N/A	<1.7	Passive	5.8-9.8	0.314	337.6	505.6
[15]	65 nm CMOS	Switched	315	5	2-15	4.095	12.9-32.8	0.04-0.10	±2.4	±0.0076	<1.27	<8.2	Passive	7.25	0.72	437.5	248.9
[19]	45nm REF	Switched	202.5	13.5	10-30	4.05	9.28-15.1	0.046-0.074	±2.91	±0.0144	<0.59	<20.2	Passive	/	0.41	493.9	753.8
					23-29	1.215	12.22-14.8	0.06-0.073	±1.29	±0.0063	<0.47	<5.4	Passive	/	0.41	493.9	206.2
This work	65 nm CMOS	Switched	150	4.84	2-19.5	2.625	12.8-20.3	0.08-0.14	±2.9	±0.0193	<1.51	<3.7	Passive	5.2-9	0.17	882.4	933.0

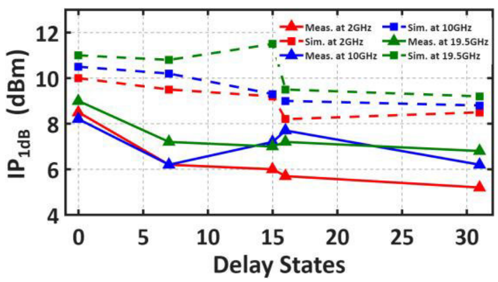


Fig. 15. Simulated and measured IP_{1dB} of the designed TTD circuit for different delay states at 2GHz, 10GHz and 19.5GHz.

from 0.08 to 0.14 dB/ps, and the loss variation per delay is within $\pm$ 0.0193 dB/ps.

IV. CONCLUSION

This brief presents a novel compact 5-bit switched TTD circuit in 65-nm CMOS technology. Operating from 2 to 19.5 GHz and occupying a compact core area of only 0.209×0.816 mm^2, the proposed design achieves a record-high area efficiency of 882.4 ps/mm^2. It provides a maximum delay tuning range of 150 ps with a high resolution of 4.838 ps across a broad bandwidth of 2-19.5 GHz. The circuit achieves an average insertion loss of 12.8-20.3 dB with a low loss variation of $\pm$ 0.0193 dB/ps. It also maintains an RMS delay error below 3.7 ps and an RMS amplitude error under 1.51 dB across the 2-19.5 GHz band.

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